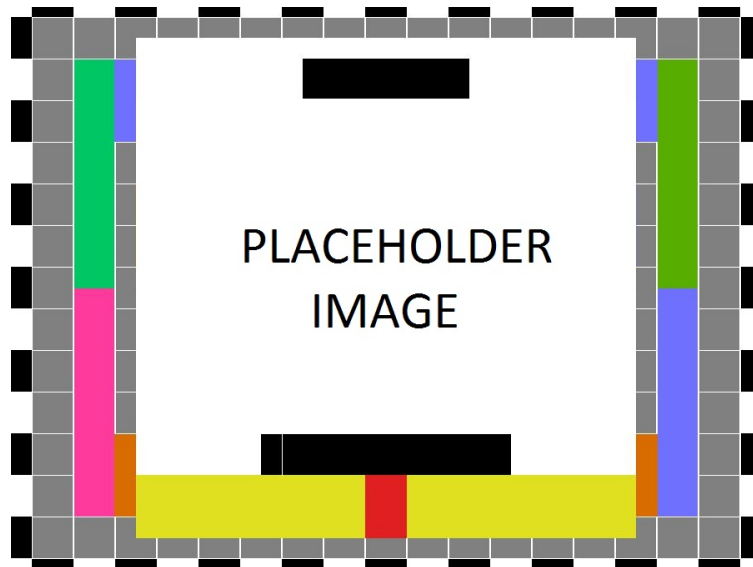


**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**DETAILED DESIGN SPECIFICATION
CSE 4317: SENIOR DESIGN II
SPRING 2027**



**TEAM FULLSTACK
CIVICPULSE COMMUNITY SERVICE-REQUEST
DASHBOARD**

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REVISION HISTORY

Revision	Date	Author(s)	Description
0.1	01.20.2027	JVN	document creation
1.0	02.05.2027	AT, GH, CB	official release

CONTENTS

1	Introduction	5
2	Design Stakeholders & Concerns	5
3	Design Viewpoints	5
4	System Overview	5
5	Presentation Layer	5
5.1	Layer Resources	5
5.2	Presentation.ResidentPortal & Presentation.StaffDashboard	5
6	Application Layer	5
6.1	Layer Resources	5
6.2	Application.APIGateway	6
6.3	Application.RequestService	6
6.4	Application.NotificationService	6
7	Data Layer	6
7.1	Layer Resources	6
7.2	Data.Datastore	6
7.3	Data.Integrations	6
8	Requirements Traceability	6
9	Appendix A	6

1 INTRODUCTION

Implementation-level design of CivicPulse Community Service-Request Dashboard as an IEEE 1016-2009 [1] Software Design Description, building on the SRS and ADS (tiers Presentation, Application, Data).

2 DESIGN STAKEHOLDERS & CONCERNS

Implementers, integrators, testers, city IT (maintainers); concerns of composition, data, interfaces, state, algorithms, and resources.

3 DESIGN VIEWPOINTS

Context, Composition, Information, Interface, Interaction & State, Algorithm, Resource per IEEE 1016; omitted where not applicable.

4 SYSTEM OVERVIEW

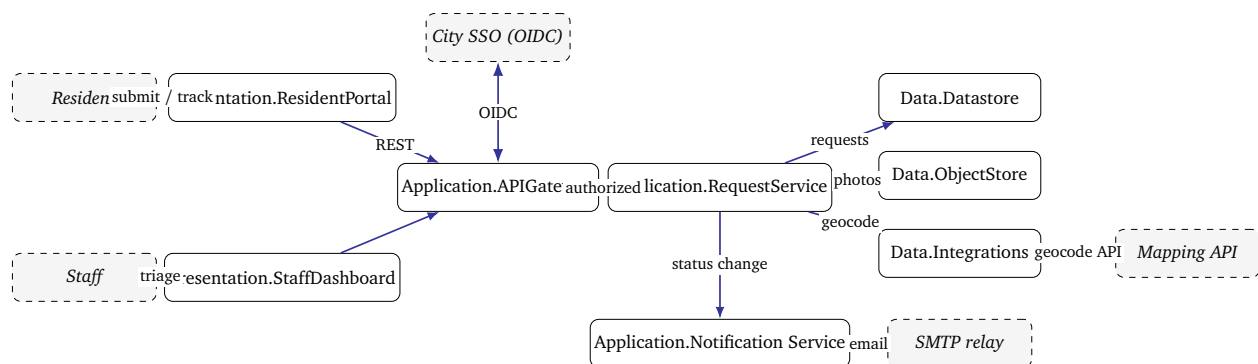


Figure 1: Three-tier deployment carried forward from the ADS

Frontend: React/TypeScript SPAs served by a CDN/reverse proxy. Backend: containerized Node/TypeScript (NestJS) API. Data: managed PostgreSQL + PostGIS and object storage.

5 PRESENTATION LAYER

5.1 LAYER RESOURCES

Deps: React, TypeScript, a map library; **Language:** TypeScript.

5.2 PRESENTATION.RESIDENTPORTAL & PRESENTATION.STAFFDASHBOARD

Composition: SPA route modules (submit, track, dashboard, request detail). **Interface (#01):** calls the REST API; renders map via EI-02. **State:** client form/query state; auth token held in memory for staff. **Interaction:** optimistic status updates with server confirmation. **Rationale:** SPA split enables responsive/accessible UI (US-01) over the stateless API (AD-01).

6 APPLICATION LAYER

6.1 LAYER RESOURCES

Deps: NestJS, an OIDC client, an ORM, an SMTP client; **Language:** TypeScript.

6.2 APPLICATION.APIGATEWAY

Interface (#02): REST; validates OIDC access tokens (EI-01) and enforces RBAC middleware (SE-01).

Algorithm: per-route role guard; deny by default.

6.3 APPLICATION.REQUESTSERVICE

Information: Request{id, category, description, geom(Point), photo_ref, status, priority, assignee, history[]}. **Interface:** CRUD + triage/assign endpoints. **State:** status machine NEW→TRIAGED→ASSIGNED→IN_PROGRESS →RESOLVED→CLOSED. **Algorithm (FR-05):** on create, ST_DWithin PostGIS query for same-category requests within radius/time; flag matches. **Rationale:** PostGIS does the spatial work natively.

6.4 APPLICATION.NOTIFICATIONSERVICE

On status change, render a template and send via SMTP (FR-04); failures retried and logged (MS-01).

7 DATA LAYER

7.1 LAYER RESOURCES

Tech: PostgreSQL + PostGIS; object storage; **Language:** SQL/migrations.

7.2 DATA.DATASTORE

Information: normalized tables requests, users, assignments, status_history, audit; spatial index on geom. **Interface:** ORM + migrations.

7.3 DATA.INTEGRATIONS

Adapters for geocoding (EI-02) and SMTP behind interfaces so providers can be swapped without touching services.

8 REQUIREMENTS TRACEABILITY

Req	Requirement (abbrev.)	Component	Iface
FR-01/02	Submit/track	Presentation.ResidentPortal, Application.RequestService	#01
FR-03	Triage/assign	Application.RequestService	#02
FR-04	Status/notify	Application.NotificationService	
FR-05	Duplicates	Application.RequestService, Data.Datastore	
SE-01	RBAC+TLS	Application.APIGateway	#02

9 APPENDIX A

Database schema, OpenAPI specification, and infrastructure-as-code are attached to the closeout package.

REFERENCES

- [1] IEEE, “IEEE Std 1016-2009 — IEEE Standard for Information Technology — Systems Design — Software Design Descriptions,” Institute of Electrical and Electronics Engineers, New York, NY, USA, Tech. Rep., 2009.